Fiscal Year:	FY 2023	**Task Last Updated:**	FY 07/10/2023
PI Name:	Limoli, Charles Ph.D.		
Project Title:	NSCOR: Mechanisms Underlying Charged Particle-Induced Disruption of CNS Function		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	climoli@uci.edu	**Fax:**	FY (949) 824-3566
PI Organization Type:	UNIVERSITY	**Phone:**	(949) 824-3053
Organization Name:	University of California		
PI Address 1:	Dept. of Radiation Oncology		
PI Address 2:	Medical Sciences I, B149		
PI Web Page:			
City:	Irvine	**State:**	CA
Zip Code:	92697-2695	**Congressional District:**	45
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2013-14 HERO NNJ13ZSA002N-NSCOR Radiation
Start Date:	07/01/2015	**End Date:**	06/30/2023
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	3
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA JSC
Contact Monitor:	Zawaski, Janice	**Contact Phone:**	
Contact Email:	janice.zawaski@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 6/30/2023 per NSSC information (Ed., 2/14/23) NOTE: End date changed to 2/28/2023 per NSSC information (Ed., 8/6/22) NOTE: End date changed to 2/28/2022 per NSSC information (Ed., 3/12/21) NOTE: End date changed to 2/28/2021 per NSSC information (Ed., 4/6/2020)		
Key Personnel Changes/Previous PI:	Per the PI, Dr. Parihar is no longer on this project and is currently working in India (Ed., 8/7/22).		
COI Name (Institution):	Baulch, Janet Ph.D. (University of California, Irvine) Britten, Richard Ph.D. (Eastern Virginia Medical School) Soltesz, Ivan Ph.D. (University of California, Irvine) Ascharya, Munjal Ph.D. (University of California, Irvine)		
Grant/Contract No.:	NNX15AI22G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Humans exposed to the space radiation environment encounter a different type of radiation than that found on Earth. Highly energetic particles can traverse the body, including the brain, and leave a path of damage that is difficult to repair and disrupts function. One consequence of this exposure is the development of radiation-induced cognitive dysfunction. The focus of this application is to resolve the mechanisms leading to radiation-induced cognitive dysfunction, in order to improve risk estimates for early and late CNS (central nervous system) effects caused by exposure to the space radiation environment. We have found that radiation impacts excitatory and inhibitory circuitry in the brain differently, and leads to significant disruptions to the structural integrity of neurons while altering the levels and activity of the proteins that regulate neuronal function. The situation is further complicated by the fact that different regions of the brain exhibit differential sensitivity to charged particle irradiation. Based on the foregoing we propose a comprehensive series of mechanistic studies analyzing the biochemical, structural, and electrophysiologic changes induced by radiation exposure that alter function to impact cognition. Carefully selected cognitive tasks and a wide variety of approaches will be used to critically analyze microcircuits in the brain and how charged particle-induced changes are transmitted via unique signaling pathways between cells that can alter gene expression profiles within cells. Collectively these studies will span a radiation effects pathway from synapse to cognition, with an overarching goal of determining whether charged particle irradiation leads to alterations in the excitatory and inhibitory activity of the brain at a global and/or region specific level. Our studies will therefore elucidate the functional consequences of radiation-induced changes on cognition that impose risk on mission critical activities and long-term cognitive health.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Work will characterize several of the adverse effects of ionizing radiation on the CNS and help to define the early and late CNS risks associated with exposure to the space radiation environment. There is a real possibility that many of the consequences of cosmic radiation exposure will extend over many years and impact (possibly negatively) neurocognitive health after return to Earth. Thus, research elucidating the mechanistic basis of charged particle-induced effects in the irradiated brain may well have functional importance to terrestrial situations where the public may be exposed to ionizing radiation from a variety of potential sources.
Task Progress:	Research Highlights for the CNS NSCOR: The sporadic nature of the finding made it challenging to initiate new experimentation. Nonetheless, we have now published our major 33-beam complex galactic cosmic radiation (GCR) simulation paper in Cellular and Molecular Life Sciences (IF 10). We have one paper under review in Nature Neuroscience and two others ready to submit. The abstracts of each of these 4 manuscripts are appended below. 1) Alaghband Y, Klein PM, Kramár EA, Cranston MN, Perry BC, Shelerud LM, Kane AE, Doan N-L, Ru N, Acharya MM, Wood MA, Sinclair DA, Dickstein D., Soltesz I, Limoli CL, Baulch JE. Galactic cosmic radiation exposure causes multifaceted neurocognitive impairments. In Press, Cell. Mol. Life Sci. USA (2022). Abstract: "Technological advancements have facilitated the implementation of realistic, terrestrial-based complex 33-beam galactic cosmic radiation simulations (GCR Sim) to now probe central nervous system functionality. This work expands considerably on prior, simplified GCR simulations, yielding new insights into responses of male and female mice exposed to 40–50 cGy acute or chronic radiations relevant to deep space travel. Results of the object in updated location task suggested that exposure to acute or chronic GCR Sim induced persistent impairments in hippocampus-dependent memory formation and reconsolidation in female mice that did not manifest robustly in irradiated male mice. Interestingly, irradiated male mice, but not females, were impaired in novel object recognition and chronically irradiated males exhibited increased aggressive behavior on the tube dominance test. Electrophysiology studies used to evaluate synaptic plasticity in the hippocampal CA1 region revealed significant reductions in long-term potentiation after each irradiation paradigm in both sexes. Interestingly, network level disruptions did not translate to altered intrinsic electrophysiological properties of CA1 pyramidal cells, whereas acute exposures caused modest drops in excitatory synaptic signaling in males. Ultrastructural analyses of CA1 synapses found smaller postsynaptic densities in larger spines of chronically exposed mice compared to controls and acutely exposed mice. Myelination was also affected by GCR Sim with acutely exposed mice exhibiting an increase in the percent of myelinated axons; however, the myelin sheathes on small calibur (< 0.3 mm) and larger (> 0.5 mm) axons were thinner when compared to controls. Present findings might have been predicted based on previous studies using single and mixed beam exposures and provide further evidence that space-relevant radiation exposures disrupt critical cognitive processes and underlying neuronal network-level plasticity, albeit not to the extent that might have been previously predicted." 2) Desai RI, Kangas BD, Luc OT, Solakidou E, Smith EC, Dawes MH, Ma X, Makriyannis A, Chatterjee S, Dayeh MA, Muñoz-Jaramillo A, Desai M, Limoli CL. Impact of complex 33-beam simulated galactic cosmic radiation exposure on touchscreen neurocognitive performance and prefrontal cortex neurotransmitter networks in mice. Submitted, Nat. Neurosci. (2023). Abstract: "Astronauts will encounter extended exposure to galactic cosmic rays (GCR) during deep space exploration, which could impair brain function and jeopardize mission success. Here, we report that neither acute nor chronic GCR exposure modified reward sensitivity in mice but did adversely affect attentional processes and increased reaction times, which could negatively impact astronauts critical time-sensitive decision-making inflight. Local potassium (K+)-stimulation in the prefrontal cortex (PFC) elevated dopamine (DA) but completely abolished temporal DA responsiveness after both acute and chronic GCR exposure, which was likely responsible for the attentional deficits. Unlike acute GCR, chronic GCR increased levels of all other neurotransmitters, with differences evident between groups after higher K+-stimulation. Correlational and machine learning analysis uncovered that acute and chronic GCR exposure differentially reorganized the connection strength and causation of DA and other monoamine PFC neurotransmitter networks compared to controls thus providing the foundation for a generalized explanation of space radiation-induced neurocognitive deficits." 3) Desorgher L, Stabilini A, Rostomyan T, Reggiani D, Hajdas W, Marcinkowski RM, Socha P, Vozenin M-C, Limoli CL, Yukihara EG, Bailat C. Dosimetry of the PM1 pion beam line at the Paul Scherrer institute for biological studies of mice. To be submitted, Radiat. Res. (2023). Abstract: "Significant past work has identified unexpected risks of central nervous system (CNS) exposure to the space radiation environment, where long-lasting functional decrements have been associated with multiple ion species delivered at low doses and dose rates. As shielding is the only established intervention capable of limiting exposure to the dangerous radiation fields in space, the recent discovery that pions, emanating from regions of enhanced shielding, can contribute significantly to the total absorbed dose on a deep space mission poses additional concerns. As a

prerequisite to biological studies evaluating pion dose equivalents for various CNS exposure scenarios of mice, a careful dosimetric validation study is required. With the goal of evaluating the functional consequences of defined pion exposures to CNS functionality, here we report on the detailed dosimetry of the PMI pion beam line at the Paul Scherrer Institute."

4) Vozenin M-C, Alaghband Y, Drayson OGG, Piaget F, Leavitt R, Allen BD, Doan N-L, Bailat C, Desorgher L, Baulch JE, Limoli CL. More may not be better: Enhanced spacecraft shielding may exacerbate cognitive decrements by increasing pion exposures during deep space exploration. To be submitted, Sci. Adv. (2023).

Abstract: "The pervasiveness of deep space radiation remains a confounding factor for the transit of humans through our solar system. Payload restrictions aside, conventional wisdom has taught us that more shielding provides enhanced protection from the harmful effects of ionizing radiation exposure. However, the fractional pion dose emanating from thicker shielded regions of the spacecraft is greater and could contribute up to 10% of the total absorbed radiation dose. Until now, the dose equivalent, or biological effect of pions on central nervous system functionality has remained untested. New results obtained with the PIM1 beam line from the Paul Scherrer Institute have revealed that low dose exposures to 150 MeV positive and negative pions (= 200 mGy), like those expected on a Mars mission, result in significant, long-lasting cognitive impairments. These surprising findings emphasize the need to carefully evaluate shielding configurations to optimize safe exposure limits for astronauts during deep space travel."

Future plans: Our pans are very straightforward over the course of the requested No-Cost Extension period (NCE). Number one priority will be to finalize all manuscripts with some remaining RNAseq analyses to perform. These activities are projected to be completed by the end of June 2023. We will also plan to reserve some funding ~$10,000 for publication-related charges and some supply funds must also be reserved for any additional follow-up studies that may be requested following peer-review. As always, we have and remain committed to helping NASA achieve its research objectives to minimize the human health and CNS risks associated with radiation exposure in deep space.

Bibliography Type:	Description: (Last Updated: 01/14/2026)
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